
Individual level analysis to look at the relationship between ASM adherence and seizures

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Format data for excel spreadsheet

Load file

```
filename = 'all_patient_diaries.xlsx';

% Extract information about number of sheets in the excel
[~, sheetNames] = xlsfinfo(filename);

% Load the data from each sheet/participant as a separate variable
for i = 1:length(sheetNames)
    sheetData = xlsread(filename, sheetNames{i});
    varName = matlab.lang.makeValidName(sheetNames{i});
    assignin('base', varName, sheetData);
end

% Extract only the seizure data (i.e., column 2) and place it all into one
% variable called seizure where each column is a different participant
for i = 1:length(sheetNames)
    sheetData = xlsread(filename, sheetNames{i});
    seizure(:,i)=sheetData(:,2);
end

% As the raw data is in half days, let's shrink the temporal data such that
% it goes by days rather than half days. The below commands just takes into
% account whether there was a seizure/no seizure for a given day. For
% example, if someone had 3 seizures in a given day, under this format they
% would just be counted as having 1 seizure day for that day.
[numRows, numCols] = size(seizure);
seizure_days = zeros(numRows / 2, numCols);
    for row = 1:numRows/2
        for col = 1:numCols
            % Check if there is a positive number in the two consecutive rows
            if seizure(2*row-1, col) > 0 || seizure(2*row, col) > 0
                seizure_days(row, col) = 1;
            end
        end
    end
```

```
end  
end  
end
```

Take into account MA

Essentially within a 90 day sliding window, this takes the number of seizure days divided by 90 for an estimate of seizure risk.

```
for i = 1:191  
    for j = 1:27  
        MA(i,j)=sum(seizure_days(i:i+89,j));  
        MA_90(i,j)=MA(i,j)/90;  
    end  
end
```

```
% Since we are looking at the relationship between MA and seizures on day  
% sliding window + 1, the first day would be day 91 since the MA requires  
% 90 days of data. Let's create a matrix of seizure days that only starts  
% at day 91 for further analysis later on.  
seizure_days_forecast=seizure_days(91:281,:);
```

Extract the ASM data for each subject

End of data format will be 3D matrix [Dose X ASM X Subject]

```
% Patient 4165: Column 3 Keppra, Column 4 Epidiolex, Column 5 Fluramine  
ASM(:,1:3,1)=BIDMC4165(:,3:5);  
  
% Patient 4216: Column 3 CBD oil, Column 4 Lamotrigine  
ASM(:,1:2,2)=BIDMC4216(:,3:4);  
  
% Patient 4251: Column 3 CBD oil, Column 5 Onfi  
ASM(:,1,3)=BIDMC4251(:,3);  
ASM(:,2,3)=BIDMC4251(:,5);  
  
% Patient 4417: Column 5 Keppra, Column 6 Lamictal XR, Column 7 Vimpat  
ASM(:,1:3,4)=BIDMC4417(:,5:7);  
  
% Patient 4614: Column 3 Aptiom, Column 4 Xcopri  
ASM(:,1:2,5)=BIDMC4614(:,3:4);  
  
% Patient 4684: Column 5 Phenobarb, Column 6 Lamotrigine, Column 8  
% Levetiracetam  
ASM(:,1:2,6)=BIDMC4684(:,5:6);  
ASM(:,3,6)=BIDMC4684(:,8);  
  
% Patient 4711: Column 3 Felbatol, Column 4 Zonisamide, Column 5 Lamictal  
ASM(:,1:3,7)=BIDMC4711(:,3:5);  
  
% Patient 4722: Column 3 Felbatol, Column 4 Onfi, Column 6 Depakote, Column  
% 7 Gabapentin, Column 8 Zonisamide, Column 9 Fenfluramine  
ASM(:,1:2,8)=BIDMC4722(:,3:4);
```

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```
ASM(:,3:6,8)=BIDMC4722(:,6:9);

% Patient 4768: Column 3 Topomax, Column 4 Epidiolex, Column 5 Lamictal,
% Column 6 Zarontin
ASM(:,1:4,9)=BIDMC4768(:,3:6);

% Patient 4835: Column 3 Oxtellar, Column 4 Epidiolex, Column 5 Clobazam,
Column 9 Fintepla
ASM(:,1:3,10)=BIDMC4835(:,3:5);
ASM(:,4,10)=BIDMC4835(:,9);

% Patient 4982: Column 3 Lamotrigine, Column 4 Banzel, Column 5 Epidiolex
ASM(:,1:3,11)=BIDMC4982(:,3:5);

% Patient 5057: Column 4 Keppra, Column 6 Onfi
ASM(:,1,12)=BIDMC5057(:,4);
ASM(:,2,12)=BIDMC5057(:,6);

% Patient 5117: Column 4 Zonisamide
ASM(:,1,13)=BIDMC5117(:,4);

% Patient 5146: Column 3 Zonisamide
ASM(:,1,14)=BIDMC5146(:,3);

% Patient 5184: Column 3 Valproic Acid, Column 4 Keppra
ASM(:,1:2,15)=BIDMC5184(:,3:4);

% Patient 5274: Column 3 Lamictal XR, Column 6 Gabapentin, Column 7
% Fycompa
ASM(:,1,16)=BIDMC5274(:,3);
ASM(:,2:3,16)=BIDMC5274(:,6:7);

% Patient 5392: Column 3 Vimpat, Column 4 Keppra, Column 5 Depakote, Column
% 6 Zonegran
ASM(:,1:4,17)=BIDMC5392(:,3:6);

% Patient 5412: Column 3 Zonegran, Column 4 Trileptal, Column 5 Fycompa
ASM(:,1:3,18)=BIDMC5412(:,3:5);

% Patient 5507: Column 3 Briviact, Column 4 Lamotrigine, Column 5
% Epidiolex, Column 6 Xcopri
ASM(:,1:4,19)=BIDMC5507(:,3:6);

% Patient 5606: Column 3 Vimpat, Column 4 Epidiolex, Column 6 Keppra,
% Column 7 Keppra XR, Column 8 XCopri
ASM(:,1:2,20)=BIDMC5606(:,3:4);
ASM(:,3:5,20)=BIDMC5606(:,6:8);

% Patient 5698: Column 3 Keppra, Column 7 Depakote ER, Column 18 XCopri
ASM(:,1,21)=BIDMC5698(:,3);
ASM(:,2,21)=BIDMC5698(:,7);
ASM(:,3,21)=BIDMC5698(:,18);

% Patient 5708: Column 9 Keppra, Column 14 XCopri, Column 20 Lamotrigine,
```

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```
% Column 21 Clobazam, Column 26 Epidiolex
ASM(:,1,22)=BIDMC5708(:,9);
ASM(:,2,22)=BIDMC5708(:,14);
ASM(:,3:4,22)=BIDMC5708(:,20:21);
ASM(:,5,22)=BIDMC5708(:,26);

% Patient 5731: Column 3 Carbamazepine, Column 4 Levetiracetam, Column 5
% Vimpat, Column 6 repeat Vimpat?? will ignore for now given no negative
% values/missed doses anyways
ASM(:,1:3,23)=BIDMC5731(:,3:5);

% Patient 5782: Column 3 Vimpat, Column 6 XCopri
ASM(:,1,24)=BIDMC5782(:,3);
ASM(:,2,24)=BIDMC5782(:,6);

% Patient 5790: Column 3 Keppra
ASM(:,1,25)=BIDMC5790(:,3);

% Patient 5799: Column 3 Keppra, Column 5 Tegretol XR
ASM(:,1,26)=BIDMC5799(:,3);
ASM(:,2,26)=BIDMC5799(:,5);

% Patient 59589: Column 3 Topamax, Column 4 Briviact, Column 5 Vimpat
ASM(:,1:3,27)=BIDMC59589(:,3:5);
```

Matrix of missed doses

The raw data for the ASM data is in half days. Let's similarly shrink the temporal resolution such that we look at ASMs in days rather than half days. We will account for missed doses in terms of "administration periods". I.e., if an individual missed morning doses of three different ASMs, that only counts as one missed administration period. For the possible range of values here, for each day it can be 0 (no missed doses), 1 (1 missed administration period, i.e., morning vs night), and 2 (missed both morning and night doses if scheduled for it).

```
[numRows, numCols, numSubs] = size(ASM);
Missed_Doses = zeros(numRows / 2, numCols, numSubs);
    for row = 1:numRows/2
        for col = 1:numCols
            for subs = 1:numSubs
                if ASM(2*row-1, col, subs) < 0 && ASM(2*row, col, subs) < 0
                    Missed_Doses(row, col, subs) = 2;
                elseif ASM(2*row-1, col, subs) < 0 && ASM(2*row, col, subs) >= 0
                    || ASM(2*row-1, col, subs) >= 0 && ASM(2*row, col, subs) < 0
                        Missed_Doses(row, col, subs) = 1;
                end
            end
        end
    end

for j = 1:27
    Storage=Missed_Doses(:, :, j);
    Missed_Doses_Periods(:, j)=max(Storage, [], 2);
end
```

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```
% Since the first 90 days are used to estimate MA, for the missed doses,
% the first number used should be on day 90 to look at how it influences
% seizure risk on day 91. The last day used should be day 280, since day
% 281 is the last day to look at seizure risk.
Missed_Doses_Periods_clean=Missed_Doses_Periods(90:280,:);

% Let's allow the creation of lag terms to look at how the effect of missed
% doses in the past influence seizure occurrence.
max_lag = 6; %in days, 6 because the data is already lagged by 1 day per
above
n_subjects = 27;

% The maximum number of predictors is max_lag + 1 (for the raw variable) + 1
(for the intercept).
B_optimal_allsubs = zeros(max_lag + 2, n_subjects);
p_allsubs = zeros(max_lag + 2, n_subjects);
ci_lower_optimal_allsubs = zeros(max_lag + 2, n_subjects);
ci_upper_optimal_allsubs = zeros(max_lag + 2, n_subjects);
chi2_p_allsubs = zeros(1, n_subjects); % Stores the chi-square p-value for
the optimal model
optimal_lags_allsubs = cell(1, n_subjects);

for i = 1:n_subjects
    % Select subject-specific data
    ASM_var = Missed_Doses_Periods_clean(:, i);
    Seizure_var = seizure_days_forecast(:, i);
    MA_var = MA_90(:, i);

    % Set X as MA or ASM, and Y as Seizure_var
    X = ASM_var;
    Y = Seizure_var;

    % --- FEATURE SCALING (Z-TRANSFORM) ---
    % Scale the predictor variable X using a z-transform (mean=0, std=1)
    mu_X = mean(X, 'omitnan');
    sigma_X = std(X, 'omitnan');

    % Check for zero variance, which indicates no missed doses for this
subject
    if sigma_X < 1e-10
        fprintf('□ Subject %d: No variability in missed doses. Skipping
model fitting.\n\n', i);
        B_optimal_allsubs(:, i) = NaN;
        p_allsubs(:, i) = NaN;
        ci_lower_optimal_allsubs(:, i) = NaN;
        ci_upper_optimal_allsubs(:, i) = NaN;
        chi2_p_allsubs(i) = NaN;
        continue; % Skip to the next subject
    end

    X_scaled = (X - mu_X) / sigma_X;

    %FIND OPTIMAL LAG COMBINATION USING AIC
    min_AIC = Inf;
```

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```
best_combo_AIC = [];  
  
fprintf('--- Subject %d: Finding Optimal Lag Combination ---\n', i);  
  
% Generate all possible lag combinations from 1 to max_lag  
lag_indices = 1:max_lag;  
valid_models_found = false;  
  
% Loop through all possible combinations of lags  
for k = 1:max_lag  
    lag_combinations = nchoosek(lag_indices, k);  
    for combo_idx = 1:size(lag_combinations, 1)  
        current_lags = lag_combinations(combo_idx, :);  
  
        % The max lag in the current combination determines the data  
trimming.  
        current_max_lag = max([0, current_lags]);  
  
        % Build the lagged predictor matrix for the current combination  
X_valid = buildLaggedMatrix(X_scaled, current_lags,  
current_max_lag);  
  
        % Trim the response variable  
y_valid = Y(current_max_lag+1:end);  
  
        % Ensure enough observations to fit the model  
if size(X_valid, 1) <= (k + 1)  
            fprintf('Skipped combination {%s}: Not enough  
observations.\n', num2str(current_lags));  
            continue;  
        end  
  
        try  
            % Fit logistic regression model  
[B, dev, stats] = glmfit(X_valid, y_valid, 'binomial',  
'link', 'logit', 'LikelihoodPenalty', 'jeffreys-prior');  
  
            % Skip if unstable  
if any(isnan(B))  
                fprintf('Skipped combination {%s}: NaN coefficients.\n',  
num2str(current_lags));  
                continue;  
            end  
  
            valid_models_found = true;  
  
            % Calculate AIC  
num_params = length(B);  
current_AIC = 2 * num_params + dev;  
  
            % Update optimal model based on AIC  
if current_AIC < min_AIC  
                min_AIC = current_AIC;  
                best_combo_AIC = current_lags;  
            end  
        catch  
            % If the model fails to fit, skip this combination  
            fprintf('Skipped combination {%s}: Model failed to fit.\n',  
num2str(current_lags));  
            continue;  
        end  
    end  
end
```

```
end

catch ME
    fprintf('Skipped combination {%s} due to error: %s\n',
num2str(current_lags), ME.message);
    continue;
end
end
end

%DETERMINE AND FIT THE OPTIMAL MODEL
if valid_models_found
    % Use the AIC-selected model.
    optimal_lags = best_combo_AIC;
    optimal_lags_allsubs{i} = optimal_lags;

    fprintf('\n □ Optimal lag combination: {%s} (AIC = %.2f)\n',
num2str(optimal_lags), min_AIC);

    % Build the final predictor matrix for the optimal model
    optimal_max_lag = max([0, optimal_lags]);
    X_opt = buildLaggedMatrix(X_scaled, optimal_lags, optimal_max_lag);
    y_opt = Y(optimal_max_lag+1:end);

    % Fit the final logistic regression model
    [B_optimal, dev_optimal, stats_optimal] = glmfit(X_opt, y_opt,
'binomial', 'link', 'logit', 'LikelihoodPenalty', 'jeffreys-prior');

    %MODEL SIGNIFICANCE TEST (CHI-SQUARE)
    [~, dev_null, ~] = glmfit(ones(size(y_opt)), y_opt, 'binomial',
'link', 'logit', 'LikelihoodPenalty', 'jeffreys-prior');
    chi2_statistic = dev_null - dev_optimal;
    df = size(X_opt, 2);
    chi2_p_value = 1 - chi2cdf(chi2_statistic, df);

    %CALCULATE CONFIDENCE INTERVALS
    se_optimal = stats_optimal.se;
    df_optimal = stats_optimal.dfe;
    alpha = 0.05;
    t_critical = tinv(1 - alpha / 2, df_optimal);
    ci_lower_optimal = B_optimal - t_critical * se_optimal;
    ci_upper_optimal = B_optimal + t_critical * se_optimal;

    % Store the results
    B_optimal_allsubs(1:length(B_optimal), i) = B_optimal;
    p_allsubs(1:length(stats_optimal.p), i) = stats_optimal.p;
    ci_lower_optimal_allsubs(1:length(ci_lower_optimal), i) =
ci_lower_optimal;
    ci_upper_optimal_allsubs(1:length(ci_upper_optimal), i) =
ci_upper_optimal;
    chi2_p_allsubs(i) = chi2_p_value;

    % Display the results
    fprintf('\nFinal Model Results for Subject %d\n', i);
```

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```
fprintf('Optimal Lags: {%s}\n', num2str(optimal_lags));
disp('Final Model Coefficients:');
disp(B_optimal');
disp('Final Model P-values:');
disp(stats_optimal.p');
fprintf('Chi-square Model p-value: %.4f\n\n', chi2_p_value);
else
    fprintf('□ No valid models could be fitted for subject %d.\n', i);
    B_optimal_allsubs(:, i) = NaN;
    p_allsubs(:, i) = NaN;
    ci_lower_optimal_allsubs(:, i) = NaN;
    ci_upper_optimal_allsubs(:, i) = NaN;
    chi2_p_allsubs(i) = NaN;
end
end

%HELPER FUNCTION: BUILD LAGGED MATRIX
function lagged_matrix = buildLaggedMatrix(X, lags, max_lag)
    % Builds a predictor matrix with the unlagged variable and selected lags.
    % All variables are trimmed to the same length.

    % Initialize matrix with the unlagged, trimmed variable
    lagged_matrix = X(max_lag+1:end);

    % Add lagged variables, each trimmed to the same length as the first.
    for l = lags
        % The start index for a lagged variable is adjusted by the maximum
        lag.
        start_idx = max_lag - l + 1;
        end_idx = length(X) - l;
        lagged_var = X(start_idx:end_idx);
        lagged_matrix = [lagged_matrix, lagged_var];
    end
end
```

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